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Editor-in-Chief
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Dear Editor,

We are pleased to submit our manuscript titled "The Measurement Protocol Decides the Conclusion: A Methodological Study of CNN and Vision Transformer Adversarial Robustness Comparisons" for consideration as a Regular Article in *IEEE Access*.

Comparisons of adversarial robustness across architecture families report conflicting conclusions, and the disagreement is usually attributed to models, training recipes, or datasets. This paper shows that a substantial part of it comes from the measurement protocol itself. Holding the models, the data, the threat model, and the attack budget all fixed, and varying only how robustness and transfer are scored, we quantify how much of a published conclusion is decided by a choice that is rarely reported. Our contributions are:

- **The absolute ordering, and what carries it.** The CNN leads under AutoAttack on the full test set ($37.93 \pm 0.14\%$ against $29.14 \pm 0.40\%$; McNemar $p < 10^{-84}$ in every run), and a conditional decomposition factors that score exactly into clean accuracy and conditional susceptibility. Both factors favor the CNN: it starts from a 12.3-point higher clean accuracy and also loses a smaller fraction of what it solves (conditional fooling 48.58% against 55.53% under PGD-10). Reporting the single number hides that attribution, which changed between our earlier checkpoints and three fresh runs while the absolute ordering did not.
- **The protocol effect, measured against the training effect.** Across four conditioning protocols, the measured CNN-to-ViT transfer asymmetry on CIFAR-10 ranges from +4.4 to +19.4 points, a mean per-seed spread of 15.01 ± 0.84 points, while retraining both architectures from scratch moves the same quantity by a standard deviation of only 0.23 to 1.27 points. We do not leave that spread as one number. It separates into 3.91 points between the two protocols we endorse, 5.31 points that the identity below corrects in closed form, and 5.79 points contributed by a protocol that estimates a different quantity. Only the first is irreducible discretion, and at 3.91 points it is still more than twice the training effect.
- **Raw transfer rates are confounded by an identity, not a tendency.** In a 3×3 matrix that adds a WideResNet-28-10 reference, the deviation of unconditioned from conditioned rates is algebraically equal to the target's clean error scaled by one minus the conditioned rate. The relation holds to within 0.41 points across 36 measured directions, which turns a known qualitative caveat into a correction applicable to already-published raw rates.
- **When the architectures are matched, the protocol decides the sign.** On SVHN, where the same recipe leaves the two models only 1.85 points apart in clean accuracy, two of

the four protocols report a CNN advantage and two report a ViT advantage. The reversal of the unconditioned rate is the arithmetic consequence of the identity above, so it can be predicted in advance.

- **Replication and threat-model transfer.** The design is repeated on CIFAR-100, where the spread stays in the same range and the sign is preserved in all twelve measurements, and re-evaluated under an L_2 budget with three pre-registered predictions, one of which fails under the reading we treat as binding and is reported as failing.
- **Three readings fail to replicate under direct measurement.** Scale-invariant gradient sparsity differences do not correspond to spatial localization differences (all four spatial measures are null); CLS attention entropy does not change measurably under attack ($n = 1000$, equivalence accepted in 12 of 12 layers); and a ViT-specific transfer attack gives no gain over plain MI-FGSM under a matched budget. Only the last is a published claim; we test the first two because our own earlier figures encouraged them.

The work closes with a short set of reporting requirements for cross-architecture robustness studies. Every number in the manuscript is produced from per-sample evaluation logs by scripts that also serve as automated consistency checks, so the protocol comparison can be reproduced without retraining.

We believe this work suits *IEEE Access* because it is methodological rather than architecture-specific: its subject is how a widely used quantity is measured, and its conclusions apply to any study comparing robustness across model families. The journal's interdisciplinary readership and open-access model fit a contribution whose value lies in being adopted as reporting practice.

Declarations:

- This manuscript is original work and has not been published elsewhere.
- The manuscript is not under consideration by any other journal.
- A preliminary subset of these results, limited to CIFAR-10, is under review as a conference paper; the present manuscript is the extended version and cites it. The conference paper reports the same three training runs per architecture; what the journal version adds is the CIFAR-100 and SVHN replications, the L_2 threat model, the three-architecture matrix and the behavioral analyses.
- All authors have reviewed and approved the final manuscript.
- All experiments use the publicly available CIFAR-10, CIFAR-100, and SVHN datasets and standard evaluation protocols.
- The authors declare no conflict of interest.
- Upon acceptance we will publish the analysis pipeline, the scripts that generate every table and figure, the per-sample evaluation records, the fixed validation-split indices, and the seed lists. Trained checkpoints total about 36 GB and are available from the authors on request.

Thank you for considering our manuscript. We look forward to your editorial decision.

Sincerely,

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